

Eugenio Sotelo

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Education

Florida Polytechnic University

Bachelors of Science in Computer Engineering – GPA: 3.53

Spring 2026

Relevant Skills: Embedded Operating systems, Autonomous Robotic Systems, VLSI Design, Kinematics and Controls of Robotic Systems, Firmware Design, Electric and Hybrid Vehicles, Digital Electronics, Electronic Motor control

Masters of Science in Electrical Engineering – GPA: 3.50

Expected Spring 2027

Experience

Polk County BOCC

June 2024 - August 2024

Student Intern

- Worked with desktop and hardware support teams to provide troubleshooting and support for customers and citizens
- Worked with mobile support team to deploy, replace, and fix county cell phones and iPads.

Phoenix Racing Solar Racing Team

October 2024 – June 2025

Battery Lead

- Designed and assembled a full high-voltage battery system for a solar-powered vehicle competing in the 2025 American Solar Challenge.
- Led custom PCB development, use of the Proheliion BMS with cell-level fusing and monitoring, to deliver a reliable and safe, competition-ready battery system.

Project Work

32-bit CPU Project

Computer Engineering Design Lab

- Designed and built a full 32-bit CPU that supports 10+ instructions as well as the ability to input data and display outputs.
- Utilized Intel Quartus Prime and the DE10-Lite to upload code and test custom ROMs

Brushless Nerf Blaster Design

Personal project

- Designed and fabricated custom PCBS integrating an RP2040, MOSFET-driven solenoid, and brushless motor control to produce a high-power, competition-ready blaster system.
- Utilized CAD modeling and 3D printing to design and manufacture components to secure electronics integration.

Solenoid Control PCB

Personal Project

- Designed a Circuit that handles the Powering and fast field collapse of the solenoid. An array of P-type and N-type MOSFETs were used in conjunction with Zener Diodes to allow the magnetic field of the solenoid to collapse.
- Utilized a modified half H-bridge design to allow the powering and collapse of the solenoids magnetic field to happen as quickly as possible.

Extracurricular Activities

Club President

Nerf-Tech

- Manage the entirety of the Club's E-board and oversaw all activities and events.
- Continue to run a per-semester HvZ, as well as plan and manage all future events.
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Skills

Software: Onshape, SolidWorks, Fusion360, Matlab, Microsoft Office, Multisim, Simulink, Arduino IDE, Microchip Studio, Overleaf, C, C++, Firmware Design

Hardware: FDM 3D Printing, DC Circuit Design, PCB Design, Soldering, Oscilloscopes, Function Generators